Yuetao Chen

+86 18912301696 chenyuetao21s@ict.ac.cn chenyuetao.tech

Education

Institute of Computing Technology, Chinese Academy of Sciences (CAS) September 2021 – June 2024
Master of Science in Computer Science
- **GPA**: 3.79 / 4.00 **Advisors**: Prof. Lei Liu & Prof. Li Chen

Beijing Jiaotong University September 2017 – June 2021
Bachelor of Engineering in Computer Science
- **GPA**: 3.93 / 4.00 **Ranking**: 1 / 54

Lancaster University September 2017 – June 2021
Bachelor of Science in Computer Science
- **GPA**: 3.81 / 4.00 **First class honours**
- 1 year at Bailrigg campus, UK; 3 years at Lancaster University College at Beijing Jiaotong University, China

Experience

Research Intern October 2022 – Present
Microsoft Research Asia *Beijing, China*

Open Source Contributor June 2022 – September 2022
OpenSUSE *Online*

Back-End Development Intern November 2017 – September 2018
Information Center, Beijing Jiaotong University *Beijing, China*

Publications

** indicates the corresponding author.*

[1] **Yuetao Chen**, Kun Li*, Yuhao Wang, Donglin Bai, Lei Wang, Lingxiao Ma, Liang Yuan, Yunquan Zhang, Ting Cao, Mao Yang, ConvStencil: Transform Stencil Computation to Matrix Multiplication on Tensor Cores, *29th ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming (PPoPP 2024)* (Accepted)

[2] Lei Liu*, Xinglei Dou, **Yuetao Chen**, An Intelligent Scheduling Approach on Mobile OS for Optimizing UI Smoothness and Power, *To be submitted to MICRO 2024*

[3] Kun Li, Zhichun Li, **Yuetao Chen**, Zixuan Wang, Yiwei Zhang, Liang Yuan, Haipeng Jia, Yunquan Zhang, Ting Cao*, Mao Yan, Tetris: Exploiting More Parallelisms for Stencil Computation on Heterogeneous Architecture via Tetrominoes Tessellation *To be submitted to SOSP 2024* [outdated version link]

[4] Yiwei Zhang, Kun Li*, Liang Yuan, **Yuetao Chen**, Haipeng Jia, Yunquan Zhang, Ting Cao, Jigsaw: Toward Conflict-free Vectorized Stencil Computation by Tessellating Swizzled Registers *To be submitted to SC 2024*

[5] Lei Liu*, Xinglei Dou, **Yuetao Chen**, Intelligent Resource Scheduling for Co-located Latency-critical Services: A Multi-Model Collaborative Learning Approach, *21st USENIX Conference on File and Storage Technologies (FAST 2023)* [link]

[6] **Yuetao Chen**, Keni Qiu, Li Chen, Haipeng Jia, Yunquan Zhang, Limin Xiao, Lei Liu*, Smart Scheduler: an Adaptive NVM-Aware Thread Scheduling Approach on NUMA Systems, *CCF Transactions on High Performance Computing* [link]

Research Projects

Transform Stencil Computation to Matrix Multiplication on Tensor Cores [1] February 2023 – Present
Advisor: Dr. Kun Li | Microsoft Research Asia *Beijing, China*
- We proposed ConvStencil, a novel stencil computing system designed to efficiently transform stencil computation to matrix multiplication on Tensor Cores.
- Proposed *stencil2row* layout transformation which significantly saves memory footprint by over 70%.
- Proposed *dual tessellation* algorithm which increases Tensor Core utilization from 12.5% to 87.5%.
- Proposed *ditry bits padding* to reduce bank conflicts and branch diverges, improving the performance by 21.4%.

Mobile Platform Resource Scheduling for Optimizing UI Smoothness and Power [2] October 2022 – Present
Advisor: Prof. Lei Liu & Prof. Li Chen | Institute of Computing Technology, CAS *Beijing, China*
- It is called MobiRL, a reinforcement learning-based resource scheduler for mobile systems.
- Implemented a resource scheduling framework for Android devices using the Deep Deterministic Policy Gradient (DDPG) model to optimize performance.

- Designed a piecewise reward function that optimizes energy consumption while limiting frame loss rate, ensuring both efficient use of resources and a high-quality user experience.
- Evaluated by the OPPO mobile system development team, our MobiRL outperforms the current scheduler by reducing 38.1% frame loss rate and 3.1% power consumption.

Stencil Computing on Heterogeneous Platform [3][4] September 2022 – Present

Advisor: Dr. Kun Li | Microsoft Research Asia *Beijing, China*

- It is the first system for high-performance stencil computation on heterogeneous CPU+GPU with novel optimizations on both CPU and GPU.
  - Implemented a new CPU SIMD register level algorithm for stencil computing, resulting in significant performance improvements of 6.3% - 19.9% on different stencil kernel shapes.
  - Referred to Tessellate Tiling (SC '17) and implemented a tiling algorithm, allowing vectorization and tiling algorithms to work together efficiently, handle boundary cases effectively, and reduce mutual interference between them.
  - Designed a data exchange method between CPU and GPU for stencil computation which covers the communication latency by computation.

Intelligent Resource Scheduling for Co-located Services [5] June 2021 – Present

Advisor: Prof. Lei Liu | Institute of Computing Technology, CAS *Beijing, China*

- It is called OSML which employs multiple ML models to work collaboratively to predict QoS variations, shepherd the scheduling, and recover from QoS violations in complicated co-location services.
  - Divided the DQN model into two parts to prevent incorrect scheduling decisions and ensure the QoS.
  - Designed a resource sharing policy, such as shared cacheline, to optimize resource utilization across co-located services.
  - Tuned PARTIES (ASPLOS '19) and CLITE (HPCA '20) for the evaluation and collected and annotated program resource allocation data.

NVM-aware thread scheduling approach on NUMA systems [6] September 2021 – May 2022

Advisor: Prof. Lei Liu & Prof. Li Chen | Institute of Computing Technology, CAS *Beijing, China*

- It is an NVM-aware thread scheduling approach for NUMA systems with hybrid memory, utilizing the LinUCB algorithm to guide scheduling decisions and reduce benchmark execution time by up to 59.9%.
  - Found the significant difference in local and remote access bandwidth between NVM and DRAM led to performance loss for existing schedulers.

Fast Fourier Transforms Library October 2020 – December 2020

Advisor: Prof. Haipeng Jia | Institute of Computing Technology, CAS *Beijing, China*

- Discovered the changing pattern of FFT algorithm with base 2.
- Automatically generated C language codes for FFT algorithm with base 2.

Development Experience

Dynamic Detection of Error Conditions from openQA Test Results June 2022 – September 2022

OpenSUSE [Issue link] [PR link] *Online*

- Communicated with the community and maintainers to refine what they want and evaluate how I did at the time.
- Added new functionality to openQA, enabling the system to search for similar failed test cases.
- Implemented visualization features in openQA to display failed tests and provide an intuitive understanding of common failures and their relationships, enabling more efficient and effective debugging.

Talk

Discovering Hot Topics of Technology News in Social Media June 2021

School of Computing and Communications, Lancaster University *Lancaster, UK*

- Invited talk about my final year project.

Awards

Merit Student	University of Chinese Academy of Sciences 2023
E Fund Management Co. Scholarship (12/420)	Institute of Computing Technology, Chinese Academy of Sciences 2023
Postgraduate Academic Scholarship	University of Chinese Academy of Sciences 2021, 2022
Excellent Bachelor Thesis	Beijing Jiaotong University 2021
First Class Academic scholarship (1/54)	Beijing Jiaotong University 2018, 2019

Technical Skills

Languages: C, Python, CUDA, C++
Tools and Framework: Linux, Git, NSight, CPU intrinsics, Intel CAT, cuDNN, Pytorch/libtorch, Django